

- Note: (i) Please bunch all your answers to Q.1 at one place (in one or more pages).
(ii) For the Q.1 parts (a) to (i), you must give the output of the code, or answer the question assuming the given code snippet has been executed.
(iii) If you think code given is grammatically incorrect, then say so. (It is called syntax error.)
(iv) If you cannot write program, but know how to solve the problem, then explain the algorithm. Due credit will be given.

(1) Attempt questions to get a maximum score of 10 out of 14.

(CO2, CO3) [10]

- (a) What will be the value of r? `int p = 8, q = 5, r = p / q * q / p;` [1]
(b) What will be the value of f? `int i; float f = i = 3.5;` [1]
(c) What will be the value of x? `int x = (9 > 7) - (7 > 7);` [1]
(d) What will be the value of g? `float g = (float)(7/5);` [1]
(e) What is the output of the following code snippet: [1]

```
void main ()
{
    int k=4;
    void func1 (int);
    func1 (k);
    printf ("%d", k);
}

void func1 (int t)
{
    t = t + 5;
}
```

(f) What is the output of the following code snippet: [1]

```
int x = 3;
switch (x)
{
    case 2: x++; break;
    case 4: x += 3; break;
    default: x += 5;
}

printf ("%d", x);
```

(g) What is the output of the following code snippet: [1]

```
int k = 4; switch (k)
{
    case 2: k--; break;
    case 4: k *= k;
    case 6: k += k;
    case 8: k++; break;
    default: k += 3;
}

printf ("%d", k);
```

(h) What is the output of the following code snippet: [2]

```
int a=5, b=3, c=4; --a < ++b ? a++ - ++b : --a + b--;
printf ("%d %d", b, c);
```

(i) What is the output of the following code snippet: [2]

```
int i = 4, j = 5, k = i++ > --j && ++i > j--;
printf ("%d %d", j, k);
```

- (j) Why is it sometimes necessary to precede an identifier name with a '&', like when using function scanf()? [1]
(k) Give any two features of the C Pre-processor. [2]

(2) (a) Briefly distinguish between primary and secondary memories in a computer. (2)

(b) "unsigned Sigma (unsigned k)" returns the sum "1 + 2 + ... + k". Draw a flow-chart to calculate Sigma (k) for a given k. (CO1) [2]
(4) (CO1) [4]

(3) Write a program to accept a coordinate point in an X-Y coordinate system and determine in which quadrant the coordinate point lies. [Hint: There are four quadrants.] (6) (CO3) [6]

(4) Write a program that prints a pattern in 10 lines, such that the first line has a '*' in the 10th column, the second line has a '*' in column 9, third in column 8, ..., the tenth line has a '*' in column 1. (3) (CO3) [6]

(5) (a) The rule for a year Y being a leap year is that "Y is divisible by 400 or (Y is divisible by 4 but not by 100)". Write a function "int isLeap (unsigned int Y)" for a program that accepts a positive integer as the year number, determines if it is a leap year or not. (6) (CO3) [6]

-OR-

(b) Write a program to display the ASCII table on the screen. Present data in two columns; program the display to enable the user to see the table without it scrolling off. [Hint: "Press any key for next page"] (CO3) [6]

(6) (a) A Perfect Number is a positive integer which equals the sum of its proper divisors, including 1. E.g., 6 equals 1 + 2 + 3, which are the proper divisors of 6. Likewise, 28 = 1 + 2 + 4 + 7 + 14. Write a program that lists all Perfect Numbers below 10000. [Hint: One way could be to code a function like "isPerfectNumber (unsigned)" that returns true / false.] (CO3) [6]

-OR-

(b) Compound interest is calculated as $A = P(1+r)^n$, where A: Amount after n periods, P: Principal, r: Rate of Interest, n: Number of periods. Taking $P = 100$ and $r = 0.1$ (10%), write a program to create a table that would display the Amount accumulated at the end of each year for the next 10 years. (4) (CO3) [6]